ARCHITECTURE AND DECISION RECORD

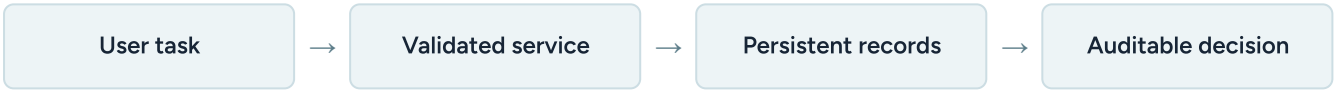
A service route planning and field-reporting portal

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React | Leaflet | PostgreSQL | Route constraints

Problem and contribution

This original design case study explores how to produce reviewable collection plans that respect explicit capacity and access constraints while recording actual service outcomes separately. I authored the requirements breakdown, architecture boundaries, data relationships, failure scenarios, and review rationale presented here. The design is a fictional presentation example, with no claim of a production deployment or measured client result.



Architecture decisions and tradeoffs

- Treat a route as a proposal subject to human review; distance alone is not the sole objective.
- Keep planned stops and actual outcomes separate so later analysis can explain what changed.
- Use synthetic pickup coordinates for the study and restrict operational location data in any later deployment.

Core records

Record	Responsibility
pickup_points	A point can appear in many dated route plans.
vehicles	A route references one vehicle and must remain within its stated capacity.
route_plans	Plans retain the proposal and approved version.
route_stops	Ordered stops separate planned service from actual stop outcomes.

The service layer owns authorization and business constraints. Interface validation improves the experience but cannot replace server checks. Event records preserve attributable decisions and their timing without logging unnecessary private content.

VERIFICATION AND REVIEW PRACTICE

Failure cases, evidence, and limitations

Proposed verification matrix

Scenario	Method	Expected behavior
Capacity constraint	Construct stops whose combined load exceeds the assigned vehicle capacity.	The plan is rejected or split with a clear explanation of the constraint.
Road change	Close a known segment after a proposal is generated but before approval.	The supervisor sees the stale constraint and must review an updated route.
Missed-stop traceability	Record a blocked pickup and schedule a follow-up collection.	Both original outcome and follow-up are preserved without appearing as two successful visits.
Baseline comparison	Plan the same synthetic stop set manually and with the proposed heuristic.	Compare feasibility, estimated distance, planning time, and documented overrides.

Technical review

Evaluate whether a route respects capacity, access, and shift limits before comparing distance. Preserve planned stops and observed outcomes separately so a later adjustment does not erase the original dispatch decision.

How to evaluate the design

Report constraint violations, estimated travel distance, estimated shift duration, planning task time, and the proportion of stops with recorded outcomes. Distinguish map estimates from measured driving distance and do not claim global optimality or real fuel savings.

The matrix specifies intended tests, not completed test results. A reviewer should capture the environment, initial records, inputs, expected state, observed state, and any unresolved defect before interpreting the outcome. Repeated runs should use separate unique request keys where relevant and preserve failures for analysis.

Scope limitations

No live vehicle tracking, public household location feeds, guaranteed shortest routes, automatic dispatch, or unsupported environmental impact claims.

Check route feasibility and the difference between a planned path and an observed collection outcome before considering route-length improvements.

> **Authorship and reuse**
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> Prepared by Gabriel Aquino as an original presentation work sample. All narrative, schemas, and test scenarios are original illustrative material. There are no external client documents, employment certificates, real personal records, or fabricated citations in this case study.